Generalized Additive Model for Council District II (2023 - 2025 Sales)

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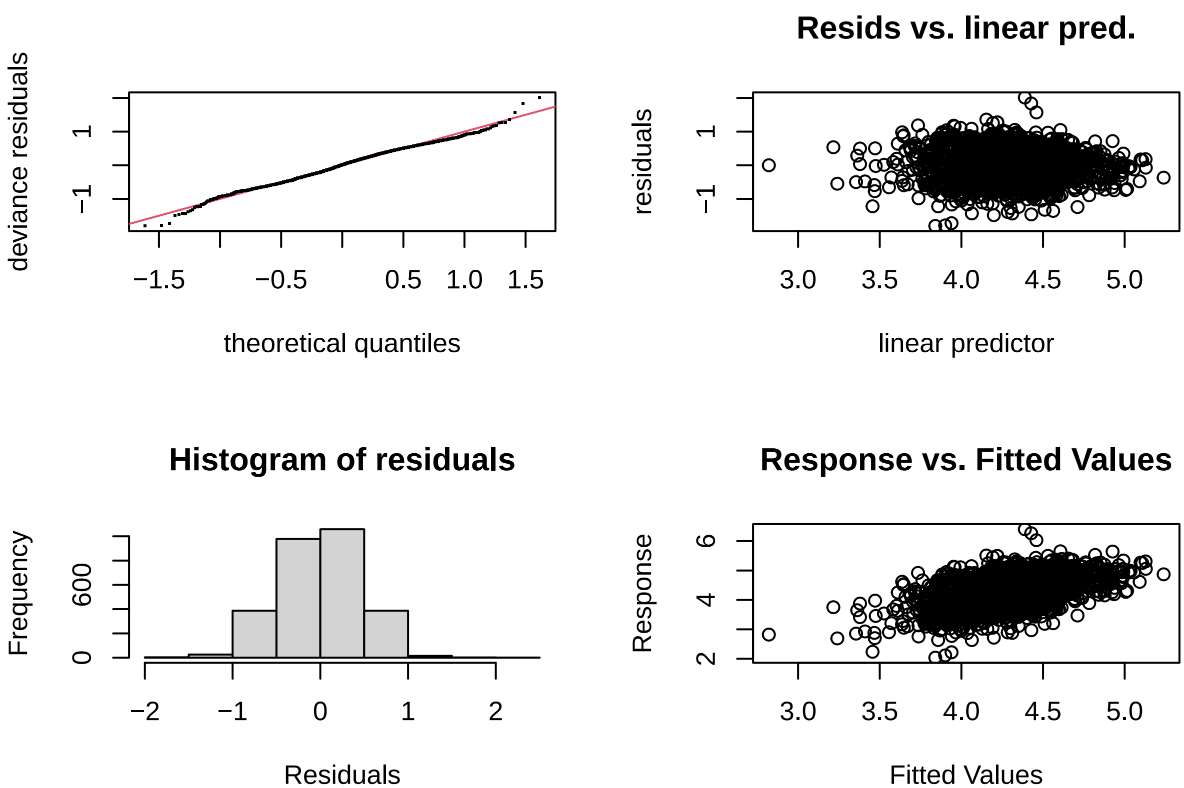
Contents

Model I

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_sppsf ~ arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          4.278708   0.032442 131.889  < 2e-16 ***
## arterial            -0.039873   0.037256  -1.070 0.284610
## single_family1story -0.019013   0.025135  -0.756 0.449455
## half_duplex1story   -0.129780   0.093090  -1.394 0.163386
## half_duplex2story   -0.077176   0.086807  -0.889 0.374052
## smallest             0.166092   0.039484   4.207 2.67e-05 ***
## small                0.080008   0.028658   2.792 0.005276 **
## large               -0.035096   0.026922  -1.304 0.192470
## largest             -0.007116   0.044470  -0.160 0.872885
## two_plus_baths       0.048805   0.051798   0.942 0.346165
## powder_room          0.003826   0.029992   0.128 0.898494
## good                 0.288265   0.069054   4.175 3.08e-05 ***
## fair                -0.171358   0.021494  -7.972 2.24e-15 ***
## poor                -0.219474   0.059282  -3.702 0.000218 ***
```

```
## very_poor_unsound   -0.667501   0.132355  -5.043 4.87e-07 ***
## qabove_avg           0.335183   0.045159   7.422 1.52e-13 ***
## qbelow_avg          -0.074410   0.025376  -2.932 0.003392 **
## crawl               -0.302367   0.132762  -2.278 0.022830 *
## slab                -0.029287   0.079121  -0.370 0.711294
## fireplace            0.139520   0.021713   6.426 1.54e-10 ***
## hot_water            0.084376   0.032228   2.618 0.008890 **
## elec_baseboard       0.513944   0.474897   1.082 0.279246
## wall                 0.180815   0.459070   0.394 0.693706
## central_air          0.064853   0.022521   2.880 0.004011 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                       edf Ref.df      F  p-value
## s(sresb_yearbuilt) 4.665      9 20.232  < 2e-16 ***
## s(ln_garagespaces) 1.267      3  3.575 0.000754 ***
## s(months)          4.081      9 11.320  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.225   Deviance explained = 23.4%
## -REML = 1843.8  Scale est. = 0.20411    n = 2857
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-9.68264e-08,1.052465e-10]
## (score 1843.835 & scale 0.204111).
## Hessian positive definite, eigenvalue range [0.5232585,1416.507].
```

```
## Model rank =  45 / 45
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                       k'  edf k-index p-value
## s(sresb_yearbuilt) 9.00 4.67    0.93  <2e-16 ***
## s(ln_garagespaces) 3.00 1.27    0.92  <2e-16 ***
## s(months)          9.00 4.08    0.93  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2857     0.000754   0.000872    0.000642  1.36  48.6 -0.929
```

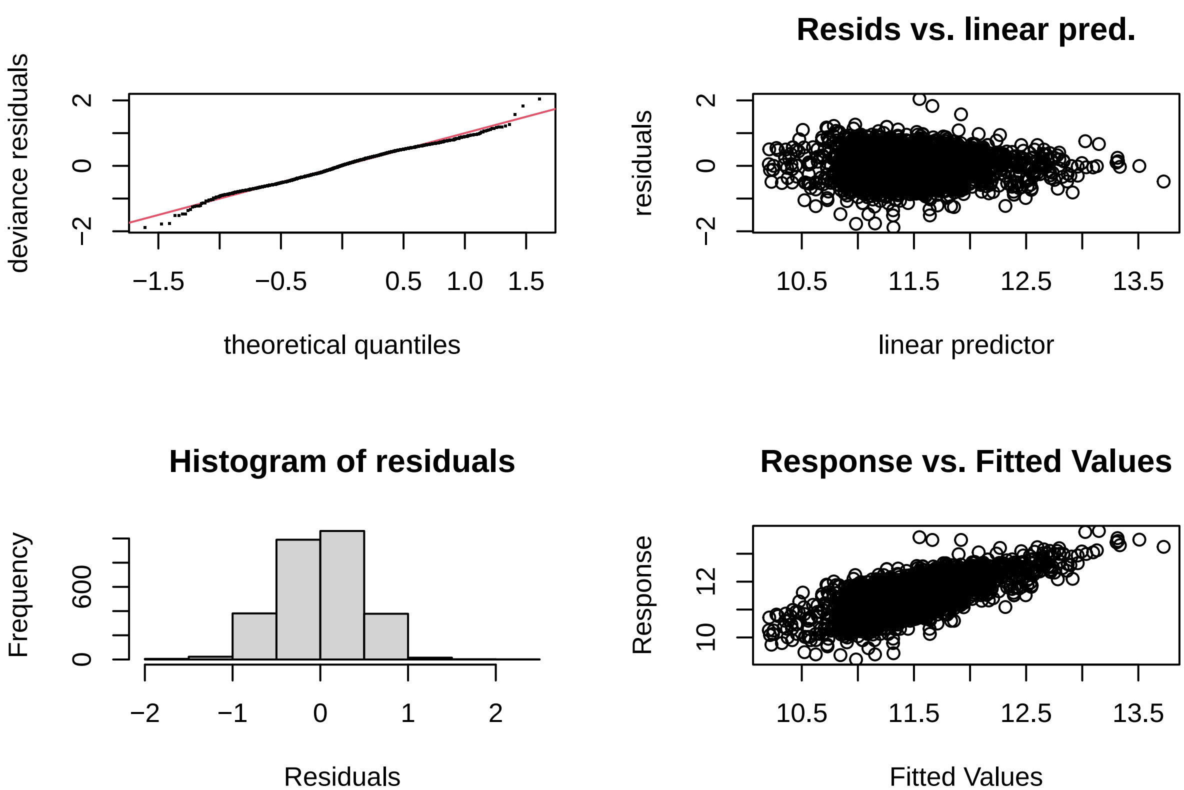
Table 1: District 2 LN_SPPSF NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2857	0.0007962	0.0009133	0.0006708	1.361417	47.80678	-0.0007231

Model II

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.421214   0.037226 306.810  < 2e-16 ***
## arterial            -0.042188   0.037200  -1.134 0.256853
## single_family1story -0.032367   0.025326  -1.278 0.201362
## half_duplex1story   -0.301239   0.096883  -3.109 0.001894 **
## half_duplex2story   -0.236733   0.091903  -2.576 0.010048 *
## smallest             0.007149   0.081360   0.088 0.929985
## small                0.005339   0.042507   0.126 0.900054
## large                0.028365   0.040016   0.709 0.478469
## largest              0.084826   0.065265   1.300 0.193804
## two_plus_baths       0.104017   0.073910   1.407 0.159432
## powder_room          0.030160   0.035193   0.857 0.391515
```

```
## good                   0.278470   0.068914   4.041 5.47e-05 ***
## fair                  -0.173542   0.021471  -8.083 9.30e-16 ***
## poor                  -0.221575   0.059118  -3.748 0.000182 ***
## very_poor_unsound     -0.662783   0.131966  -5.022 5.42e-07 ***
## qabove_avg             0.319446   0.046295   6.900 6.39e-12 ***
## qbelow_avg            -0.093196   0.025790  -3.614 0.000307 ***
## crawl                 -0.308315   0.133072  -2.317 0.020580 *
## slab                  -0.056409   0.079672  -0.708 0.478996
## fireplace              0.137928   0.021845   6.314 3.15e-10 ***
## hot_water              0.083741   0.032311   2.592 0.009598 **
## elec_baseboard         0.497598   0.473684   1.050 0.293586
## wall                   0.205267   0.457747   0.448 0.653878
## central_air            0.059813   0.022506   2.658 0.007914 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                              edf Ref.df      F  p-value
## s(ln_landratio)            1.457      9  1.556 0.000114 ***
## s(ln_base_bldgsize_ratio) 2.316      9  3.900  < 2e-16 ***
## s(sresb_yearbuilt)         4.802      9 16.725  < 2e-16 ***
## s(ln_garagespaces)         1.176      3  2.400 0.005202 **
## s(months)                  4.156      9 11.806  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.471   Deviance explained = 47.8%
## -REML = 1839.5  Scale est. = 0.20284    n = 2857
```




```
##
```

```
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-1.99168e-05,1.407614e-05]
## (score 1839.537 & scale 0.2028395).
## Hessian positive definite, eigenvalue range [0.3940744,1416.509].
## Model rank =  63 / 63
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                              k'  edf k-index p-value
## s(ln_landratio)            9.00 1.46    0.95    0.01 **
## s(ln_base_bldgsize_ratio) 9.00 2.32    0.93  <2e-16 ***
## s(sresb_yearbuilt)        9.00 4.80    0.94  <2e-16 ***
## s(ln_garagespaces)        3.00 1.18    0.92  <2e-16 ***
## s(months)                 9.00 4.16    0.93  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2857        0.941       1.06       0.873  1.21  40.6 -0.504
```

Table 2: District 2 LN_PRICE (No ECFs) NLM I Assessing Coefficients

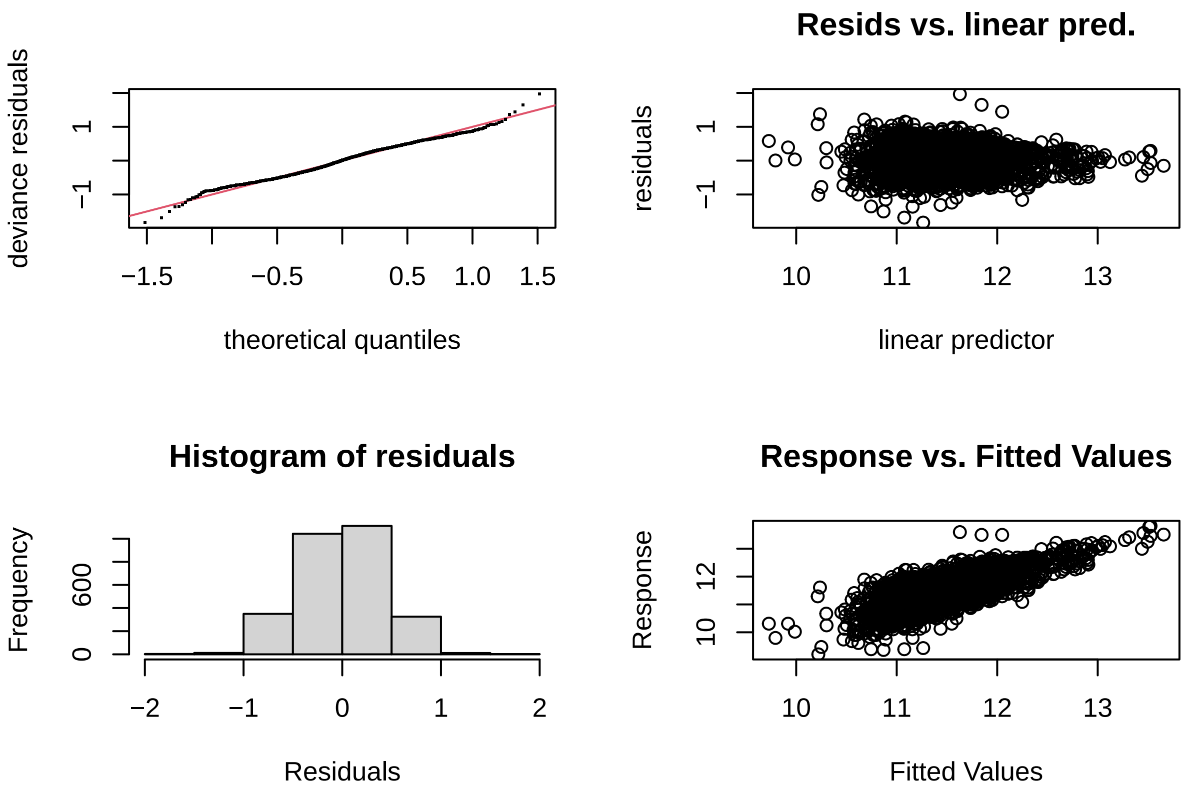
n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2857	0.982639	1.10695	0.9125807	1.212989	40.28099	-0.6114004

Model III

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10) + ecf
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.305275   0.048018 235.438  < 2e-16 ***
## arterial            -0.048059   0.038216  -1.258 0.208655
## single_family1story -0.013844   0.024158  -0.573 0.566643
## half_duplex1story   -0.215919   0.092080  -2.345 0.019101 *
```

```
## half_duplex2story   -0.149471   0.086737  -1.723 0.084949 .
## smallest            -0.019628   0.071726  -0.274 0.784369
## small               -0.005958   0.038497  -0.155 0.877013
## large                0.055675   0.036381   1.530 0.126046
## largest              0.131850   0.060304   2.186 0.028868 *
## two_plus_baths       0.073445   0.068706   1.069 0.285170
## powder_room          0.004102   0.032892   0.125 0.900764
## good                 0.245236   0.065816   3.726 0.000198 ***
## fair                -0.176962   0.020416  -8.668  < 2e-16 ***
## poor                -0.183428   0.055955  -3.278 0.001058 **
## very_poor_unsound   -0.614748   0.124803  -4.926 8.89e-07 ***
## qabove_avg           0.123535   0.048416   2.552 0.010778 *
## qbelow_avg          -0.045249   0.028377  -1.595 0.110929
## crawl               -0.293709   0.125543  -2.340 0.019379 *
## slab                -0.060813   0.076299  -0.797 0.425496
## fireplace            0.080722   0.020697   3.900 9.84e-05 ***
## hot_water            0.049527   0.029554   1.676 0.093882 .
## elec_baseboard       0.823640   0.448855   1.835 0.066615 .
## wall                 0.187633   0.430948   0.435 0.663308
## central_air          0.019258   0.021438   0.898 0.369106
## ecf2R202             0.171962   0.062368   2.757 0.005868 **
## ecf2R203             0.235219   0.044758   5.255 1.59e-07 ***
## ecf2R204             0.391362   0.130038   3.010 0.002639 **
## ecf2R205             0.093066   0.042510   2.189 0.028661 *
## ecf2R206             0.421279   0.304011   1.386 0.165939
## ecf2R207             0.253777   0.072674   3.492 0.000487 ***
## ecf2R208             0.800695   0.095899   8.349  < 2e-16 ***
## ecf2R209             0.613121   0.166965   3.672 0.000245 ***
## ecf2R210             1.197629   0.241495   4.959 7.50e-07 ***
## ecf2R211            -0.900283   0.196434  -4.583 4.78e-06 ***
## ecf2R212             0.111429   0.039342   2.832 0.004654 **
## ecf2R213             0.204083   0.044087   4.629 3.84e-06 ***
## ecf2R214             0.356601   0.042105   8.469  < 2e-16 ***
## ecf2R215             0.641347   0.093388   6.868 8.02e-12 ***
## ecf2R216             0.800686   0.232279   3.447 0.000575 ***
## ecf2R217             0.230913   0.315151   0.733 0.463799
## ecf2R218            -0.264912   0.095916  -2.762 0.005784 **
## ecf2R219             0.539909   0.135152   3.995 6.64e-05 ***
## ecf2R220             0.024164   0.051743   0.467 0.640539
## ecf2R221             0.096887   0.049132   1.972 0.048709 *
## ecf2R222            -0.107125   0.049147  -2.180 0.029363 *
## ecf2R223            -0.010325   0.051599  -0.200 0.841412
## ecf2R224            -0.228846   0.053861  -4.249 2.22e-05 ***
## ecf2R225            -0.038859   0.053146  -0.731 0.464729
## ecf2R226            -0.363433   0.120886  -3.006 0.002667 **
## ecf2R227             0.344612   0.139790   2.465 0.013753 *
## ecf2R228            -0.600290   0.135538  -4.429 9.83e-06 ***
## ecf2R229            -0.343025   0.135915  -2.524 0.011664 *
## ecf2R230            -0.600152   0.249244  -2.408 0.016110 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                              edf Ref.df      F  p-value
## s(ln_landratio)            1.338      9  0.930 0.003007 **
## s(ln_base_bldgsize_ratio) 1.639      9  1.780 4.64e-05 ***
## s(sresb_yearbuilt)        1.850      9  2.709 1.70e-06 ***
## s(ln_garagespaces)        1.328      3  4.129 0.000327 ***
## s(months)                 4.648      9 12.093  < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.532   Deviance explained = 54.2%
## -REML = 1689.9  Scale est. = 0.17973   n = 2857
```



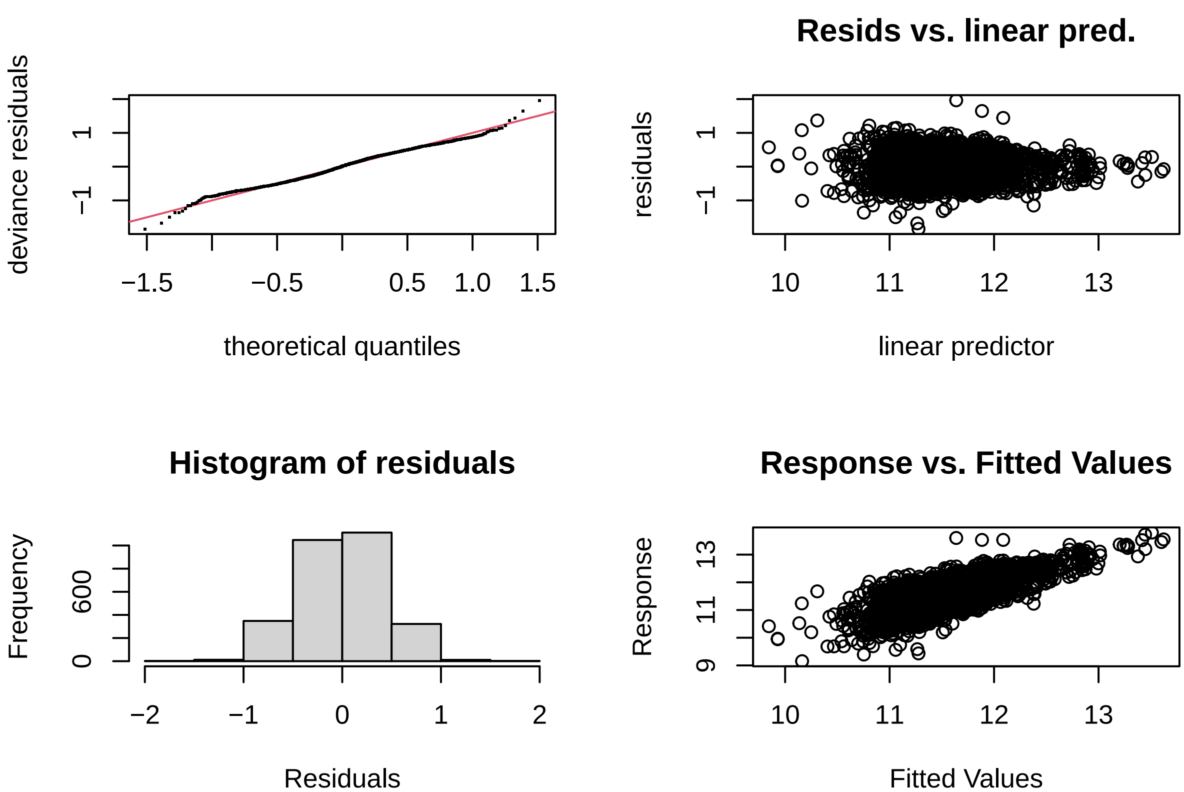

```
##
## Method: REML   Optimizer: outer newton
## full convergence after 7 iterations.
## Gradient range [-3.473006e-05,2.690509e-05]
## (score 1689.949 & scale 0.1797282).
## Hessian positive definite, eigenvalue range [0.3922048,1402.006].
## Model rank =  92 / 92
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                              k'  edf k-index p-value
## s(ln_landratio)            9.00 1.34    1.01    0.76
## s(ln_base_bldgsize_ratio) 9.00 1.64    0.97    0.07 .
## s(sresb_yearbuilt)        9.00 1.85    0.99    0.30
## s(ln_garagespaces)        3.00 1.33    0.99    0.31
## s(months)                 9.00 4.65    0.98    0.09 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2857        0.945       1.04       0.886  1.18  37.7 -0.408
```

Model IV

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_tasp ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + ecf + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         11.356100   0.046901 242.131  < 2e-16 ***
## arterial            -0.048157   0.038168  -1.262 0.207159
## single_family1story -0.016918   0.021070  -0.803 0.422084
## half_duplex1story   -0.218165   0.090621  -2.407 0.016128 *
## half_duplex2story   -0.145721   0.086483  -1.685 0.092106 .
## smallest            -0.019244   0.071563  -0.269 0.788021
## small               -0.006477   0.038419  -0.169 0.866128
## large                0.055478   0.036294   1.529 0.126486
## largest              0.131587   0.060177   2.187 0.028849 *
## two_plus_baths       0.074078   0.068633   1.079 0.280526
## powder_room          0.003750   0.032853   0.114 0.909131
## good                 0.247584   0.065663   3.770 0.000166 ***
## fair                -0.177813   0.020177  -8.813  < 2e-16 ***
## poor                -0.184102   0.055787  -3.300 0.000979 ***
## very_poor_unsound   -0.620054   0.124006  -5.000 6.08e-07 ***
## qabove_avg           0.122878   0.048387   2.539 0.011156 *
## qbelow_avg          -0.045077   0.028360  -1.589 0.112071
## crawl               -0.293239   0.125482  -2.337 0.019515 *
## slab                -0.061972   0.076120  -0.814 0.415635
## fireplace            0.080920   0.020679   3.913 9.33e-05 ***
## hot_water            0.050103   0.029508   1.698 0.089633 .
## elec_baseboard       0.829434   0.448415   1.850 0.064462 .
## wall                 0.180398   0.430560   0.419 0.675260
## central_air          0.020128   0.021398   0.941 0.346959
## ecf2R202             0.172607   0.062234   2.774 0.005582 **
## ecf2R203             0.235188   0.044734   5.257 1.57e-07 ***
## ecf2R204             0.393645   0.129855   3.031 0.002456 **
## ecf2R205             0.093376   0.042455   2.199 0.027929 *
## ecf2R206             0.427998   0.303775   1.409 0.158967
## ecf2R207             0.252991   0.072539   3.488 0.000495 ***
## ecf2R208             0.801110   0.095716   8.370  < 2e-16 ***
## ecf2R209             0.612414   0.166876   3.670 0.000247 ***
## ecf2R210             1.194488   0.240972   4.957 7.59e-07 ***
## ecf2R211            -0.897032   0.196161  -4.573 5.02e-06 ***
## ecf2R212             0.111336   0.039302   2.833 0.004647 **
## ecf2R213             0.204464   0.044043   4.642 3.60e-06 ***
## ecf2R214             0.356779   0.042057   8.483  < 2e-16 ***
## ecf2R215             0.640415   0.093313   6.863 8.27e-12 ***
```

```
## ecf2R216               0.803223   0.232080   3.461 0.000546 ***
## ecf2R217               0.218720   0.313243   0.698 0.485083
## ecf2R218              -0.262293   0.095787  -2.738 0.006215 **
## ecf2R219               0.537292   0.134958   3.981 7.03e-05 ***
## ecf2R220               0.023520   0.051672   0.455 0.649018
## ecf2R221               0.097315   0.049061   1.984 0.047403 *
## ecf2R222              -0.105829   0.049090  -2.156 0.031184 *
## ecf2R223              -0.009750   0.051545  -0.189 0.849981
## ecf2R224              -0.228092   0.053830  -4.237 2.34e-05 ***
## ecf2R225              -0.039043   0.053110  -0.735 0.462322
## ecf2R226              -0.363275   0.120642  -3.011 0.002626 **
## ecf2R227               0.347229   0.139633   2.487 0.012950 *
## ecf2R228              -0.597355   0.135398  -4.412 1.06e-05 ***
## ecf2R229              -0.343993   0.135842  -2.532 0.011386 *
## ecf2R230              -0.599535   0.249103  -2.407 0.016158 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                                edf Ref.df     F  p-value
## s(ln_landratio)           1.336444      9 0.938 0.002865 **
## s(ln_base_bldgsize_ratio) 1.636296      9 1.776 4.73e-05 ***
## s(sresb_yearbuilt)        1.852151      9 2.733 1.54e-06 ***
## s(ln_garagespaces)        1.331619      3 4.173 0.000307 ***
## s(months)                 0.002036      9 0.000 0.769545
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.522   Deviance explained = 53.1%
## -REML = 1681.3  Scale est. = 0.17963   n = 2857
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 8 iterations.
## Gradient range [-0.0008546313,0.0001632999]
## (score 1681.33 & scale 0.1796303).
## Hessian positive definite, eigenvalue range [0.0008534331,1402.002].
## Model rank =  92 / 92
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                                k'     edf k-index p-value
## s(ln_landratio)           9.00000 1.33644    1.01   0.700
## s(ln_base_bldgsize_ratio) 9.00000 1.63630    0.97   0.060 .
## s(sresb_yearbuilt)        9.00000 1.85215    0.99   0.320
## s(ln_garagespaces)        3.00000 1.33162    0.99   0.320
## s(months)                 9.00000 0.00204    0.98   0.085 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2857        0.987       1.09       0.926  1.18  37.4 -0.426
```